

Mehanske karakteristike:

Beton: C40/50

$$f_{ck} = 4,0 \text{ kN/cm}^2$$

$$\gamma_c = 1,5$$

$$f_{cd} = 2,67 \text{ kN/cm}^2$$

$$E_{cm} = 3500 \text{ kN/cm}^2$$

$$f_{ctm} = 0,35 \text{ kN/cm}^2$$

Armatura: B500B

$$f_{yk} = 50 \text{ kN/cm}^2$$

$$\gamma_s = 1,15$$

$$f_{yd} = 43,48 \text{ kN/cm}^2$$

$$E_s = 20000 \text{ kN/cm}^2$$

Geometrijske karakteristike:

$$b = 30 \text{ cm}$$

$$h = 60 \text{ cm}$$

$$\text{Krovni sloj: } 30 \text{ mm}$$

$$d_g = 16 \text{ mm}$$

$$a = 4,8 \text{ cm}$$

$$d = 55,2 \text{ cm}$$

Obremenitev:

$$M_{Ed} = 258,07 \text{ kNm}$$

$$N_{Ed} = 0 \text{ kN}$$

$$z_s = 25 \text{ cm}$$

$$M_{Eds} = 258,1 \text{ kNm}$$

Potrebna vzdolžna armatura:

$$k_d = 0,106$$

$$k_{s,Rogač} = 1,068$$

$$k_{s,EC2} = 1,061$$

$$A_{s,potr.} = 11,48 \text{ cm}^2$$

Minimalna vzdolžna armatura:

$$A_{s,min} = 3,01 \text{ cm}^2$$

Minimalna razdalja med sosednjima palicam

$$e_{min} = 21 \text{ mm}$$

Maksimalno število palic v eni vrsti:

$$n_{max} = 6,51$$

Izberemo:

$$n = 6$$

$$\phi_{vzd} = 16 \text{ mm}$$

$$\phi_{streme} = 10 \text{ mm}$$

$$A_{s,dej.} = 12,06 \text{ cm}^2$$

$$e = 4,4 \text{ cm}$$

Pravokotni prerez - enojna armatura											Tabela 2	
ε_s [%]		$-\varepsilon_1$ [%]										
		1.00	1.25	1.50	1.75	2.00	2.25	2.50	2.75	3.00	3.25	3.50
1.5	k_x	.400	.455	.500	.538	.571	.600	.625	.647	.667	.684	.700
	k_s	1.163	1.193	1.220	1.247	1.273	1.298	1.323	1.347	1.370	1.391	1.411
	k_h	.143	.189	.230	.268	.299	.325	.346	.364	.379	.391	.402
3.0	k_x	.250	.294	.333	.368	.400	.429	.455	.478	.500	.520	.538
	k_s	1.096	1.117	1.137	1.157	1.176	1.196	1.216	1.235	1.254	1.272	1.289
	k_h	.095	.130	.165	.197	.227	.252	.274	.293	.310	.325	.338
4.0	k_x	.200	.238	.273	.304	.333	.360	.385	.407	.429	.448	.467
	k_s	1.075	1.092	1.109	1.126	1.143	1.160	1.177	1.194	1.210	1.226	1.241
	k_h	.078	.108	.138	.168	.194	.218	.240	.259	.276	.291	.304
5.0	k_x	.167	.200	.231	.259	.286	.310	.333	.355	.375	.394	.412
	k_s	1.062	1.076	1.091	1.105	1.120	1.135	1.150	1.165	1.179	1.193	1.207
	k_h	.065	.092	.119	.145	.170	.192	.213	.231	.247	.262	.276
6.0	k_x	.143	.172	.200	.226	.250	.273	.294	.314	.333	.351	.368
	k_s	1.053	1.065	1.078	1.091	1.103	1.117	1.130	1.143	1.156	1.169	1.181
	k_h	.057	.080	.104	.128	.151	.172	.191	.208	.224	.239	.253
7.0	k_x	.125	.152	.176	.200	.222	.243	.263	.282	.300	.317	.333
	k_s	1.046	1.057	1.068	1.079	1.091	1.103	1.115	1.127	1.138	1.150	1.161
	k_h	.050	.071	.093	.115	.136	.155	.173	.190	.205	.219	.232
8.0	k_x	.111	.135	.158	.179	.200	.220	.238	.256	.273	.289	.304
	k_s	1.040	1.050	1.060	1.071	1.081	1.092	1.103	1.113	1.124	1.135	1.145
	k_h	.044	.064	.084	.104	.123	.141	.158	.174	.189	.202	.215
9.0	k_x	.100	.122	.143	.163	.182	.200	.217	.234	.250	.265	.280
	k_s	1.036	1.045	1.054	1.064	1.073	1.083	1.093	1.103	1.113	1.122	1.132
	k_h	.040	.058	.076	.095	.113	.130	.146	.161	.175	.188	.200
10.0	k_x	.091	.111	.130	.149	.167	.184	.200	.216	.231	.245	.259
	k_s	1.033	1.041	1.049	1.058	1.067	1.076	1.085	1.094	1.103	1.112	1.121
	k_h	.037	.053	.070	.087	.104	.120	.135	.149	.163	.175	.187

Lastnost	Trdnostni razredi I						
Oznaka	C12/15	C16/20	C20/25	C25/30	C30/37	C35/45	C40/50
fck [MPa]	12	16	20	25	30	35	40
fck, cube [MPa]	15	20	25	30	37	45	50
fcm [MPa]	20	24	28	33	38	43	48
fctm [MPa]	1,6	1,9	2,2	2,6	2,9	3,2	3,5
fctk, 0,05 [MPa]	1,1	1,3	1,5	1,8	2	2,2	2,5
fctk, 0,95 [MPa]	2	2,5	2,9	3,3	3,8	4,2	4,6
Ecm [GPa]	27	29	30	31	33	34	35
ϵ_{c1} [‰]	1,8	1,9	2	2,1	2,2	2,25	2,3
ϵ_{cu1} [‰]				3,5			
ϵ_{c2} [‰]				2			
ϵ_{cu2} [‰]				3,5			
n				2			
ϵ_{c3} [‰]				1,75			
ϵ_{cu3} [‰]				3,5			

betona					
C50/60	C55/67	C60/75	C70/85	C80/95	C90/105
50	55	60	70	80	90
60	67	75	85	95	105
58	63	68	78	88	98
4,1	4,2	4,4	4,6	4,8	5
2,9	3	3,1	3,2	3,4	3,5
5,3	5,5	5,7	6	6,3	6,6
37	38	39	41	42	44
2,45	2,5	2,6	2,7	2,8	2,8
	3,2	3	2,8	2,8	2,8
	2,2	2,3	2,4	2,5	2,6
	3,1	2,9	2,7	2,6	2,6
	1,75	1,6	1,45	1,4	1,4
	1,8	1,9	2	2,2	2,3
	3,1	2,9	2,7	2,6	2,6